

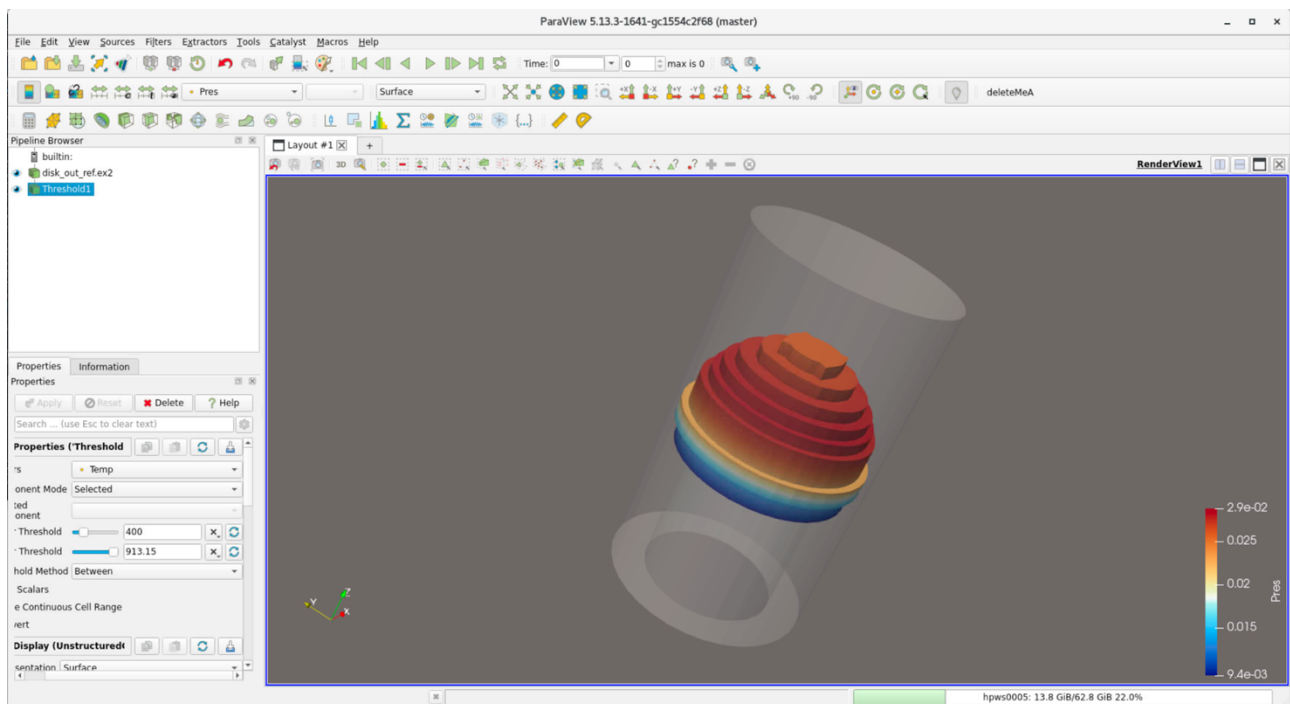


Tutorial 02 – Paraview IsoSurfaces and Contouring

<https://docs.paraview.org/en/latest/Tutorials/ClassroomTutorials/beginningSourcesAndFilters.html>

1. Threshold filter

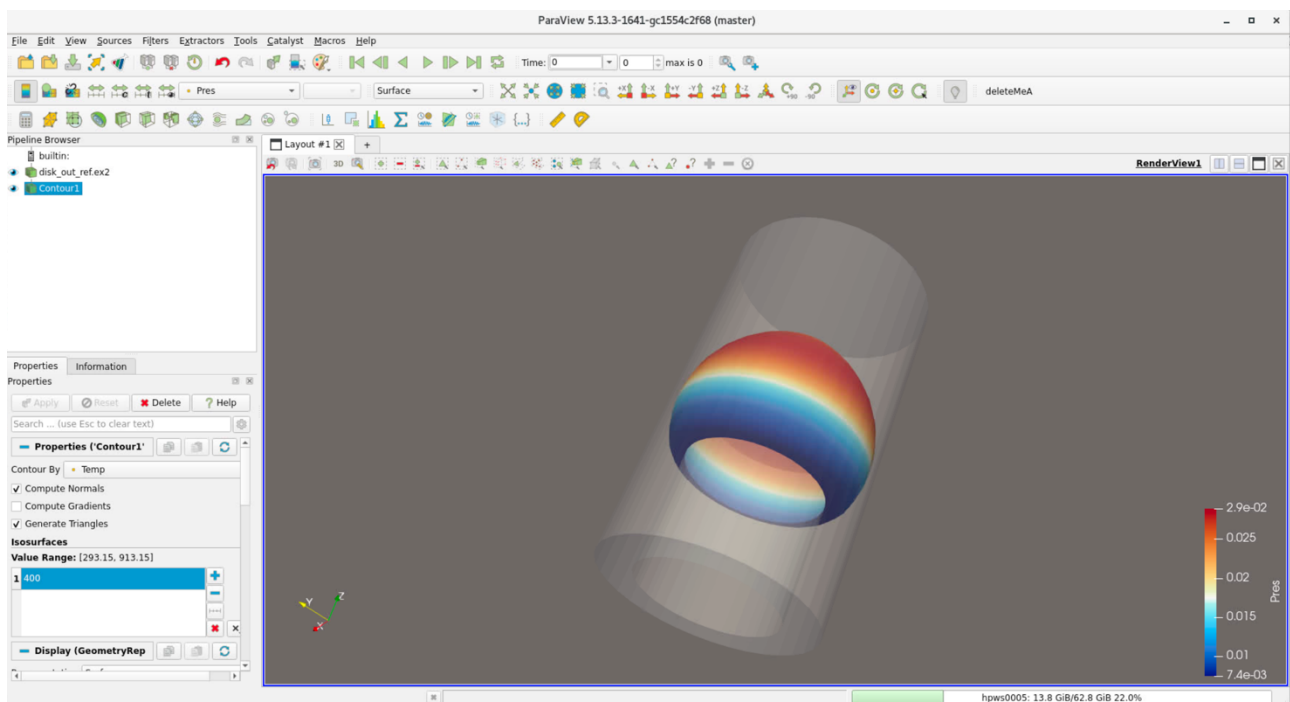
- Open disk_out_ref.ex2.
- **Apply.**
- Spin the object around, and look inside of it.
- Press the **Threshold** button , found to the far left. (Notice that you can also find this through the filters menu.)
- Select **Scalars** “Temp” (Temperature).
 - Note: The blue recycle button will set the min and max values.
- Change the **Lower Threshold** to 400.
- **Apply.**
- Set **Coloring** to **Pres.**
- Spin the object around, and look at it.
- Now, let’s place this hot section back into the cylinder.
- Let’s open another version of disk_out_ref.ex2, using the **File → Open** menu.
- **Apply.**
- Make sure that the second disk_out_ref.ex2 is highlighted in the **Pipeline Browser**
- Set **Representation** to **Wireframe**.
- In the **Properties** tab, set **Coloring** to **Pres**
- What we have done: We have created an unstructured grid holding the cells that fit our criteria. This can make our data much bigger, and should be avoided if we are working with big data.
- Extra:
 - Change the outside cylinder to be volume rendered, set **Representation** to **Volume**.



2. Contour filter

- Open disk_out_ref.ex2.
- **Apply.**
- Select **Filters** → **Common** → **Contour.**
- Change **Properties: Contour By:** to **Temp.**
- Under **Value Range**, press **Delete All.**
- Now press **New Value** and enter **400.**
- **Apply.**
- Set **Coloring** to **Temp.**
- Why are all parts of the object the same color?
- Set **Coloring** to **Pres** (Pressure).
- This represents the location inside of the cylinder that is at temperature 400, and is colored by pressure.
- Turn the visibility back on for the disk_out_ref.ex2.
- In the **Pipeline Browser**, click disk_out_ref.ex2 (turning it white)
- Select **Representation Wireframe.**
- Set **Coloring** to **Pres.**
- Notice that this is another way to see two representations of the same object by reading the object in once and modifying it. In the Threshold Filter (above), we read in the object twice, and displayed each object differently.

- Under **Contour** set **Representation** to **Surface with Edges**. What are the triangles you can see?
- What we have done: We have created an isosurface of a specific temperature. One nice thing about isosurfaces is that they decrease the amount of data that has to fit into memory. This is handy when you are displaying big data.
- Extra 1:
 - Highlight disk_out_ref.ex2.
 - Select **Clip** filter.
 - Select **X Normal** for the clip plane.
- Extra 2:
 - Under **contour**, use **New Range** to create 2 new surfaces for temp = 300 and 500.
 - Set **Coloring** to **Temp**.
 - Next, select **Clip** filter to cut the cylinder in half.
 - What are the surfaces showing us? What are the colors showing us?
 - Highlighting clip
 - Under **Properties** tab, change the **Opacity** to **.50**.



3. Clip to Scalar filter

- Open disk_out_ref.ex2.
- **Apply.**
- Select **Filters** → **Recent** → **Clip**.
- **Apply.**
- Select **Clip Type** → **Scalar**.
- Select **Scalars** → **Temp**.
- Input a **Value** of **400**.
- **Apply.**
- Select **Clip** filter again.
- Unclick **Show Plane**.
- **Apply.**
- Set **Coloring** to **Temp**.
- Highlight disk_out_ref.ex2, turn the eyeball on.
- Display by **Volume**
- Set **Coloring** to **Temp**.
- Select **Clip** filter again and play with the clipping value (Pressing **Apply** when appropriate)